

Rosacea Therapy and Classification

Rosacea therapy may include barrier protection practices, topical antimicrobials, oral antibiotics, retinoids, intense pulsed light, and vascular laser modalities for adequate long-term control of symptoms.

The Plewig-Kligman classification system describes rosacea progression through stages, with stage III corresponding to phymatous changes. In contrast, the National Rosacea Society (NRS) classification states that rosacea does not necessarily progress from one sub-type to another, but that sub-types may overlap in the same individual. A provisional grading system was developed by the NRS Expert Committee to standardize clinical assessment. Rosacea severity evaluations should also consider the psychological, social, and occupational impact of the condition, as well as individual treatment responsiveness.

Epidemiology

Although the exact prevalence of rosacea is unknown, it is considered common, particularly in Caucasian populations. While the majority of cases occur in fair-skinned individuals, rosacea can also affect people of African and Asian descent. The NRS estimates that rosacea affects approximately 14 million Americans. It occurs in both men and women, with onset typically after age 30. However, cases have been reported in children, adolescents, and young adults.

Etiology and Pathogenesis

The clinical variability among rosacea sub-types suggests possible differences in etiology and pathophysiology. These may involve facial vascular reactivity, dermal connective tissue structure or composition, extracellular matrix changes, pilosebaceous unit abnormalities, microbial colonization, or a combination of factors that alter the skin's response to triggers.

Rosacea is unmasked or exacerbated by chronic, repeated exposure to triggers—particularly those that induce flushing. Common triggers include: - Temperature extremes (hot or cold) - Sunlight - Wind - Hot beverages - Exercise - Spicy foods - Alcohol - Emotional stress - Cosmetics and topical irritants - Menopausal flushing - Flushing-inducing medications

Flush reactions are mediated by both neural and humoral mechanisms and are typically confined to the face. This facial predominance is attributed to higher baseline facial blood flow and a more superficial, extensive cutaneous vascular network compared to other body areas.

Histologic studies show dermal matrix degeneration and endothelial damage in rosacea. Contributing factors may include increased vascular permeability, impaired clearance of inflammatory mediators, or photodamaged connective tissue altering vascular and lymphatic support in the dermis. These changes can lead to chronic inflammation and persistent erythema on facial convexities in susceptible individuals.

Sun exposure is considered a contributing factor, though not sufficient alone. Solar elastosis is often present in rosacea-affected skin. However, rosacea is not more prevalent in outdoor workers, non-facial sun-damaged skin does not develop rosacea features, and photoprovocation studies do not show increased sensitivity to acute UV exposure in rosacea patients.

The mechanism of action of oral and topical antimicrobials in rosacea—whether anti-inflammatory or antimicrobial—remains debated. The role of commensal microbes such as *Propionibacterium acnes* and *Demodex folliculorum* in triggering folliculocentric inflammation is controversial. It is unclear if these

organisms directly cause inflammatory papules or if a hypersensitivity reaction to the microbes or associated bacteria (e.g., *Bacillus oleronius*) is involved.

Evidence supporting a microbe-driven mechanism in papulopustular rosacea (PPR) includes: - Poor response of rosacea lesions to corticosteroids and nonsteroidal anti-inflammatory drugs - Superior efficacy of oral tetracyclines - Effectiveness of benzoyl peroxide in patients who tolerate it

It remains uncertain whether clinical improvement in PPR requires a quantitative reduction in *P. acnes*.

Clinical Features

Erythematotelangiectatic rosacea (ETR) is characterized by: - Persistent facial erythema - Flushing - Telangiectasia - Central facial edema - Burning and stinging sensations - Skin roughness or scaling

These signs and symptoms may occur in various combinations. ETR is classified into mild, moderate, and severe sub-types based on clinical presentation.